

User guide

Source code address

<https://github.com/arvin1367/3-D-Deformation-Inversion>

Example address

<https://data.mendeley.com/datasets/nj7956xtbm/1>

How do I get set up?

You should copy the source code folder to MATLAB 2022a or higher. We provide real-time scripts (main_3d_glacier.mlx, main_3d_landslide.mlx, main_3d_land_subsidence.mlx) for different research subjects. You will have a clear understanding of how the software operates.

Usage

We have provided an example of a glacier from (Dem Glacier Data) to download the data. Here is the real-time script for the glacier, which is the usage steps:

Software and system preparation: Supports Windows, Mac, and Linux systems, requiring Matlab 2022a or higher. You should add this code package to the path first. The path contains a parent folder and five sub folders, as well as a time table of image pairs for ascending and descending. For example:

- ascending_azimuth (dir)
- ascending_range (dir)
- descending_azimuth (dir)
- descending_range (dir)
- supporting_documents (dir)
- date_ascending.xls (table)
- date_descending.xls (table)

Notice: Please ensure that the folder and file names are consistent, except for the sbas-pot processing results, which are the files in the first four folders.

Supporting_documents include the local evaluation angle (lv_theta.tif) and orientation angle (lv_phi.tif) obtained from GAMMA software (look_vector), dem_ddew.tif and dem_ddns.tif obtained from myDEMdiff(). lv_theta_a.tif and lv_theta_d.tif are ascending and descending, respectively. lv_phi_a.tif and lv_phi_d.tif are ascending and descending, respectively. An example is as follows:

```
clear  
clc
```

- (1) To calculate the first order difference between the east-west and north-south directions of the DEM, it is necessary to provide larger DEM data and reference data than the research area.

Final obtained dem_ddew.tif and dem_ddns.tif, place it in supporting_documents folder.

```
myDEMdiff('other/njbw_dem.tif','njbw/ascending_range/20221004_20221016r.tif')
```

- (2) dirc is the parent folder, this example is 'njbw'.

```
dirc='njbw';
```

- (3) Start data preprocessing, mainly including reading data, constructing zero mask, and calculating time. If you do not have a local incidence angle file, you can enter the incidence angle of ascending in the second parameter, the azimuth of ascending in the third parameter, the incidence angle of descending in the fourth parameter, and the azimuth of descending in the fifth parameter.

eg. obj_arvin=myPreprocess(dirc,'POT',36.9156,347.2974,37.8561,192.6945);

```
obj_arvin=myPreprocess(dirc,'POT');  
save('obj_arvin.mat');
```

- (4) Start data processing, mainly including constructing coefficient matrices and constant matrices, conducting 3-D inversion, saving data (TIF and MAT formats, speed, cumulative, and annual average), and calculating norms. Including two methods for solving ill conditioned equations: Tikhonov and TSVD.

For Tikhonov: obj_data=myProcess(obj_arvin,'Tikhonov',0.1,1); The second parameter is the regularization parameter, recommended 0.1 (large deformation, e.g. glacier), and the third parameter is the regularization order, usually 0, 1, 2 (recommended 1).

For TSVD: obj_data=myProcess(obj_arvin,'TSVD',0.5); The second parameter is the singular value truncation value, recommended 0.5 (large deformation, e.g. glacier).

```
obj_data=myProcess(obj_arvin,'TSVD',0.5);  
save('obj_data.mat');
```

- (5) Start the post-processing process, interactively select a stable region (draw a box in the stable region) as the random error, and then calculate the uncertainty. If an error is reported, it indicates that there are a large number of empty values in the stable region and a new selection is needed.

```
obj_un=myReprocess(obj_arvin,obj_data);
```

```
save('obj_un.mat');
```

(6) Draw deformation maps in various directions.

```
mkfigure(obj_arvin,'dem_t1_05/vu.tif','other/demkk.tif',2,'other/njbw.shp');
```

(7) Draw dynamic deformation maps in various directions.

```
mkgifs('njbw_results/results_velocity/ve','other/demkk.tif',2,[-1.5 1.5], 've.gif','other/njbw.shp');
```

(8) Draw a deformation rate map with vector arrows, with the base color representing the upper and lower shape variables, and the arrow length representing the horizontal shape variables.

```
mkfigure_v('njbw_results/vn_mask.tif','njbw_results/ve_mask.tif','njbw_results/vu_mask.tif','other/demkk.tif',2,10,10,'other/njbw.shp');
```

(9) Draw a vector arrow profile. Interactive selection of the profile to be drawn, supporting a straight line.

```
mksec('njbw_results/vn_mask.tif','njbw_results/ve_mask.tif','njbw_results/vu_mask.tif','other/demkk.tif',0.05);
```

(10) Draw a point timing chart. Interactive selection of points and drawing of their long-term cumulative deformation map.

```
mkpt('njbw_results/myFile_out',obj_arvin.date.time)
```

Function

myDEMdiff(parameter 1,parameter 2): The function is used to calculate the first-order difference of DEM in east-west (dem_ddew.tif) and north-south (dem_ddns.tif) directions for SPF constraint. Input parameter 1: DEM with slightly larger than the DEM of the research area; Input parameter 2: The size and geographic information (.tif) that needs to be output. Output: dem_ddew.tif and dem_ddns.tif.

obj = myPreprocess(parameter 1,parameter 2,varargin): The function is used for data preprocessing. Input parameter 1: The name of the parent folder where the input data is stored. Input parameter 2: The method ('POT' or 'D-InSAR') used for SAR data processing. varargin: If you do not have a local incidence angle file, you can enter the incidence angle of ascending in the second parameter, the azimuth of ascending in the third parameter, the incidence angle of descending in the fourth parameter, and the azimuth of descending in the fifth parameter. obj: struct. It includes input, time, and mask data.

obj = myProcess(parameter 1,parameter 2): The function is used to start 3D inversion. Input parameter 1: Preprocessed struct. Input parameter 2: The method ('Tikhonov' or 'TSVD')

used for ill-conditioned equation solution. If 'Tikhonov', the second parameter is the regularization parameter, and the third parameter is the regularization order, usually 0, 1, 2. If 'TSVD', the second parameter is the singular value truncation value. obj: struct. It includes the corresponding .tif and .mat for the results.

obj = myReprocess(parameter 1,parameter 2): The function is used for further statistical analysis of results. Input parameter 1: Preprocessed struct. Input parameter 2: Processed struct. obj: struct. It includes the uncertainty of results and statistical charts.

mifigure(parameter 1,parameter 2, parameter 3,parameter 4, parameter 5): The function is used to draw deformation maps in various directions. Input parameter 1: Preprocessed struct. Input parameter 2: 3-D inversion results in a certain direction. Input parameter 3: DEM of the area, which needs to be consistent with the deformation range. If not, you can use 'no' instead. Input parameter 4: Scale factor, set according to the user's drawing range. Input parameter 5: Select input, if any, it is the line shp file of the deformation contour.

mkgifs(parameter 1,parameter 2, parameter 3,parameter 4, parameter 5): The function is used to draw dynamic deformation maps in various directions. Input parameter 1: 3-D inversion rate results in a certain direction between each time period. Input parameter 2: DEM of the area, which needs to be consistent with the deformation range. If not, you can use 'no' instead. Input parameter 3: Scale factor, set according to the user's drawing range. Input parameter 4: Color bar range, set according to the deformation size. Input parameter 6: Output the name of the gif. Input parameter 6: Select input, if any, it is the line shp file of the deformation contour.

mkfigure_v(parameter 1,parameter 2, parameter 3,parameter 4, parameter 5, parameter 6,parameter 7, parameter 8): The function is used to draw a deformation rate map with vector arrows, with the base color representing the upper and lower shape variables, and the arrow length representing the horizontal shape variables. Input parameter 1: 3-D inversion results in north direction. Input parameter 2: 3-D inversion results in east direction. Input parameter 3: 3-D inversion results in vertical direction. Input parameter 4: DEM of the area, which needs to be consistent with the deformation range. If not, you can use 'no' instead. Input parameter 5: Scale factor, set according to the user's drawing range. Input parameter 6: Vector arrows use spacing, reflecting arrow density. Input parameter 7: Arrow scaling factor.

Input parameter 8: Select input, if any, it is the line shp file of the deformation contour.

mksec(parameter 1,parameter 2, parameter 3,parameter 4, parameter 5): The function is used to draw a vector arrow profile. Interactive selection of the profile to be drawn, supporting a straight line. Input parameter 1: 3-D inversion results in north direction. Input parameter 2: 3-D inversion results in east direction. Input parameter 3: 3-D inversion results in vertical direction. Input parameter 4: DEM of the area, which needs to be consistent with the deformation range. If not, you can use 'no' instead. Input parameter 5: Arrow scale.

mkpt(parameter 1,parameter 2, parameter 3,parameter 4, parameter 5): The function is used to draw a point timing diagram. Interactive selection of points and drawing of their long-term cumulative deformation map. Input parameter 1: Results(.mat). Input parameter 2: Time in the struct.